

CVP8R: IR1001-G2 Wired SD-WAN with LTE as Last Resort in Routed Mode

Overview

The single-router IR1001-G2 SD-WAN configuration catalog item assists with rapid deployments at small to medium sized locations where flexibility, simplicity and standardization are key. This catalog represents a Cisco Validated Profile (CVP) suited for deployments at locations like retail, Oil & Gas Pipelines, Roadways & Intersections.

This network configuration is wired transport with single LTE connectivity as last resort. This configuration offers network segmentation via 1 secure VPN (VPN 10)

Configuration Catalog Details

Ethernet LAN connectivity is mapped into one VPN for security and granular control:

- Connected Devices (VPN 10): Secure network or DIA for corporate devices.
- One or more Hubs are required to be deployed as shown. Hubs need to be deployed using a configuration group not provided here as part of this catalog entry. The hubs will facilitate communication between connected devices behind edge routers and the enterprise network by advertising to edge devices a default route or a list of subnets the edge devices need to reach in the enterprise. It will be up to the Hub to allow also for edge traffic to reach the internet or block it if not needed at the edge. It is also necessary to assign each edge device a unique Service VPN subnet and IP address to allow for return traffic.

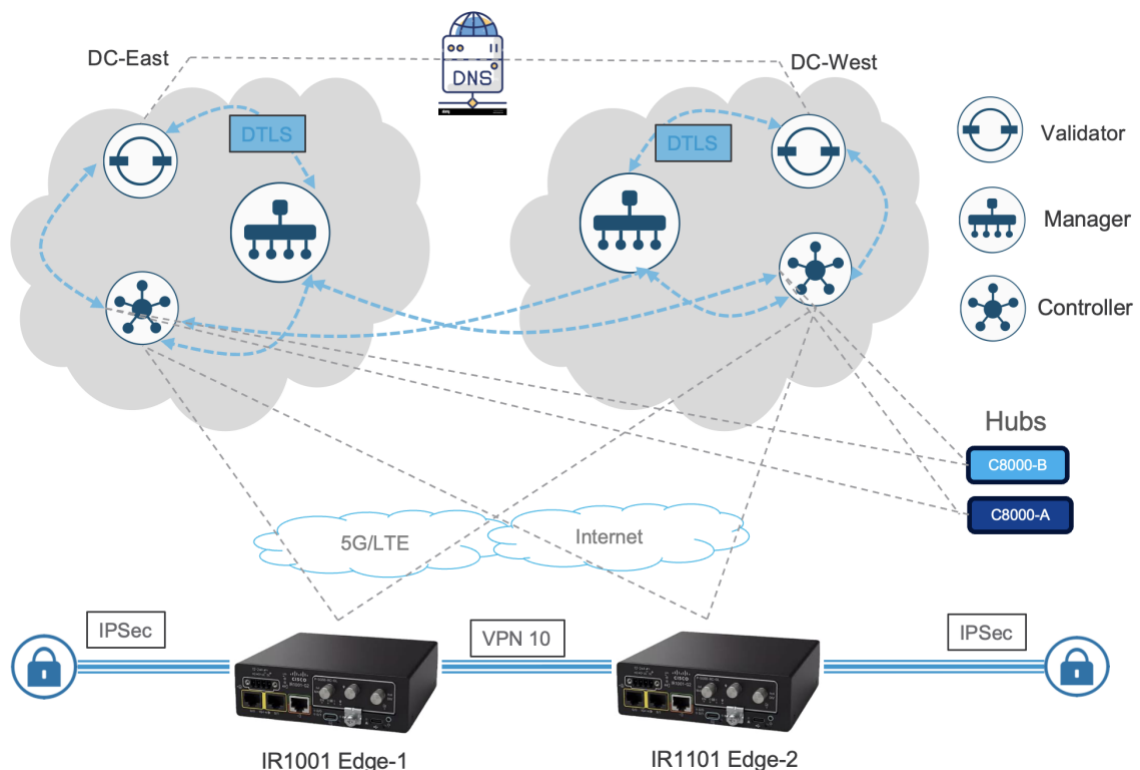


Fig 1: Configuration Catalog Design

Further details on functionality provided by this catalog entry:

- The Service VPN (LAN) side is configured to advertise static and connected routes (VPN 10) or flexible integrations at sites with routed LAN access layer. User will need to enter unique subnet for each edge device at deployment time to route back from a Hub router to each edge. User will also need to advertise all subnets from the hub router to edge devices, including a default route as needed. Below is an image of the Service section and the items user needs to set per device (in red). This can be automated for a list of devices using a csv file. If user does not want to have a DHCP server in the service VPN, the “VPN1 SVI1 DHCP Server Network”, Mask and Default gateway can be set to dummy values to prevent any connected device from having DHCP service or access through DHCP.

CVD1r_Service (service) 5 Optional

Service_VPN1 (lan/vpn)

VPN1 Secondary DNS Server: 208.67.220.222

VPN1 Primary DNS Server: 208.67.222.222

VPN1 Number: 10

vpn1_svi1 (lan/vpn/interface/svi)

VPN1 SVI1 IP Address: This field is required

VPN1 SVI1 Subnet Mask: vpn1_svi1_ip_mask* This field is required

VPN1 SVI1 Interface Name: Vlan10

vpn1_svi1_dhcp_server (dhcp-server)

VPN1 SVI1 DHCP Server Net...: vpn1_svi1_dhcp_mas... This field is required

VPN1 SVI1 DHCP Server Net...: This field is required

VPN1 SVI1 DHCP Server DNS...: Enter values separated by ','

VPN1 SVI1 DHCP Server Exc...: Enter values separated by ','

VPN1 SVI1 DHCP Server Gat...:

VPN1_Loopback (lan/vpn/interface/ethernet)

VPN1 Loopback IP: This field is required

VPN1 Loopback subnet mask: 255.255.255.255

- A DHCP Server is configured for VPN 10 allowing connected clients to receive IP addresses from service VPN 10. In addition, a few timers for BFD and OMP have been tuned to reduce LTE bandwidth usage while maintaining SD-WAN functionality. DHCP details (subnet, default route, DNS...etc) should be added as needed at deploy time (no defaults provided).
- A primary and secondary NTP servers can be configured with primary as preferred source (defaults provided).
- User can set the router console rate and set the GPS coordinates of its location (defaults provided).
- Default DNS servers (Cisco DNS) are provided in both VPN 0 and VPN 10. User can change those in VPN 10 as needed.

- User can point router logs to a server of their choice and provide the correct VPN and source interface for such traffic. The field is in the system section called “Logging Server IP Address” and is currently set to “0.0.0.0”, please change it at install time to a valid logging server IP address or set it to be the same as the “System IP” from the system section if one does not exist.
- At device deploy time, user needs to assign a loopback IP address to the field “VPN1 Loopback IP” under the service section. This can be assigned same IP address as “System IP” address and serves as a troubleshooting IP that can be advertised to and reachable from a Hub router over the same VPN.

Cisco provides the configurations below as is for your convenience. These configurations have been built using industry best practices, observed across multiple deployments, which may be beneficial to you. Cisco is not responsible for any technical issues, bugs, or other issues that may arise from your use of these configurations and any resulting indirect, incidental, reliance, consequential, special or exemplary damages or loss of actual or anticipated revenue, profit, business, savings, data goodwill or use, business interruption, damaged data, wasted expenditure or delay in delivery (in all cases, whether direct or indirect).